原题扫描整理

2026-2027 学年初二上学期质量监测试题

英语 11 页 · 物理 8 页 · 数学 6 页

按科目及原页码排序，共 25 个有效页面。

保留原照片的题目、图形及手写痕迹。

物理第 8 页重复照片已去重，采用较清晰的一张。

这是照片汇编，不是去除笔迹后的空白试卷。

答案与解析、数学辅助线图另见配套 PDF。

嘉祥教育集团2026-2027学年上学期初二质量监测试题

英 语

注意事项：

1. 全卷分为第A卷和第B卷，A卷满分100分，B卷满分50分；考试时间120分钟。
2. 考生使用答题卡作答。
3. 在作答前，考生务必将自己的姓名、考号和座位号涂写在试卷和答题卡规定的地方。考试结束，监考人员只将答题卡收回，试卷请考生自己妥善保存。
4. 选择题部分必须用2B铅笔填涂；非选择题部分必须使用0.5毫米黑色墨水签字笔书写，字体工整、笔迹清楚。
5. 请按照题号在答题卡上各题目对应的答题区域内作答，超出答题区域书写的答案无效；在草稿纸、试卷上答题均无效。
6. 保持答题卡清洁，不得折叠、污染、破损等。

A 卷

第一部分 听力（共 25 小题；计 30 分）

一、听句子，选择正确答语。每小题念两遍。（共 5 小题；每小题 1 分，计 5 分）

1.	A. It was fantastic!	B. I'm fine, thanks.	C. You too.
2.	A. Good luck!	B. Have a good trip.	C. Sounds great.
3.	A. No problem.	B. Me, too.	C. Sorry to hear that.
4.	A. Of course.	B. You are welcome.	C. Good job.
5.	A. How do you know?	B. Exactly!	C. I like plants.

二、听句子，选择与内容相符的图片。每小题念两遍。（共5小题；每小题1分，计5分）

A　　B　　C　　D　　E

6. _____　7. _____　8. _____　9. _____　10. _____

三、听对话，选择正确答案。每小题念两遍。（共10小题；每小题1.5分，计15分）

听第 1 段对话，回答第 11、12 题。

11. What are the two speakers talking about?
A. The man's friend　　B. The man's vacation　　C. The man's favorite activities

12. How long did the man stay in Hong Kong?
A. For 16 days.　　B. For 10 days.　　C. For 6 days.

听第 2 段对话，回答第 13 、14 题。

13. What is Helen going to do first?

A. Hang out with Lucy. B. Clean the bathroom. C. Sweep the floor.

14. Why does Helen want to buy flowers?

A. Grandma likes them. B. Lucy is coming. C. Dad asks her to.

听第 3 段对话，回答第 15 至 17 题。

15. When did Ted come to Beijing?

A. Last September. B. Last October. C. Last November.

16. Why does Ted like basketball best?

A. It helps him relax. B. It keeps him busy. C. It teaches him new skills.

17. What makes Ted feel at home in Beijing?

A. The local food. B. His good friends. C. The sports he plays.

听第 4 段对话，回答第 18 至 20 题。

18. What was special about white elephants in ancient Thailand?

A. They worked hard. B. They were smart.

C. They were symbols of good luck and power.

19. Why might the king give someone a white elephant?

A. To share animal care. B. To train the animal.

C. To make the person spend money.

20. Which of the following is a white elephant?

A. Something dirty but useful. B. Something expensive but useless.

C. Something interesting but expensive.

四、听短文，完成表格信息（共5小题；每小题1分，计5分）

Three Ways to Help the Bees on Our School __21__	
Why	Fewer bees than last year.
First	Plant more __22__.
Second	Give them water in a small dish. Bees stand on some __23__ in it when they drink.
Third	Be friendly. A bee hurts people only when it feels __24__.
Time for club meeting	Every __25__ afternoon.

21. club 22. flowers 23. water 24. angry 25. Thursday

第二部分 基础知识运用（共 20 小题；计 40 分）

五、阅读下面短文，从方框内的选项中选出可以填入空白处的最佳选项。(共5小题;每小题2分,计10分)

A. such vacations	B. find peace	C. looking for	D. fantastic moments	E. bring joy

"Lying-flat" (躺平) travel is a new way to relax by slowing down during holidays. Now, many young people choose to travel this way. Cheng Yi is one of them.

Cheng, a 36-year-old worker, enjoys travelling. In the past, she and her family were always on the go during holidays. They rushed to busy cities, __26__ more places of interest. That always made them tired. Now, their holidays are more about spending __27__ together as a family.

"I used to think of __28__ as a waste of time, but now they fit our family perfectly. They also __29__ to us," said Cheng.

More people are choosing "lying-flat" travel as a way to get away from busy routines and __30__. However, this doesn't mean they will stop exploring wonderful places. They are still open to other travel ideas.

六、补全对话 根据对话内容，从方框中选出适当的选项补全对话。（共5小题；每小题2分，计10分）

A: You're back, Zhang Nan. __31__.

B: It was interesting. I went to Gulangyu Island in Xiamen—a beautiful place by the sea.

A: Sounds nice! __32__

B: Yes. And my grandpa also went there with us. We all enjoyed the trip. __33__. Look! This is for you.

A: Oh, thank you! It's so kind of you!

B: __34__

A: No. I just stayed at home.

B: __35__.

A: Of course not. I read books and fed dogs. It was interesting.

A. Didn't you feel bored?
B. How was your vacation?
C. I bought many gifts for my friends.
D. Did you go there with your parents?
E. Did you go anywhere special, Li Na?

七、完形填空。阅读下面短文，根据短文内容，从A、B、C三个选项中选出可以填入空白处的最佳选项。（共10小题,每小题2分，计20分）

A country mouse loved his home, but his cousin from the town had a different opinion. One day, the town mouse came to __36__ him. The country mouse welcomed his cousin and shared his bread with him.

The town mouse thought the food was ___37___. It was not good enough for him. He talked about his cakes at home. He wanted his ___38___ to come with him. The country mouse agreed, and they travelled to the town together.

At the house, they found cakes on the table. The country mouse smiled. He was ___39___ to see so much delicious food. He reached for a cake, ___40___ a loud noise stopped him. Two dogs were outside the door. The ___41___ ran into the room, and the mice quickly ran away.

Now the country mouse was ___42___. The cakes still looked good, but he could not enjoy them. He wanted to ___43___ instead of having dinner there. His cousin was surprised and asked him to stay longer.

The country mouse said ___44___. He cared more about a quiet life than fine food. His own home had less to eat, but it gave him ___45___. With that thought, he left the town and went back to his little house.

36. A. visit	B. teach	C. follow
37. A. delicious	B. poor	C. wonderful
38. A. teacher	B. neighbour	C. cousin
39. A. disappointed	B. happy	C. sad
40. A. but	B. so	C. because
41. A. children	B. people	C. animals
42. A. angry	B. bored	C. afraid
43. A. sleep	B. leave	C. play
44. A. yes	B. hello	C. no
45. A. peace	B. strength	C. trouble

第三部分 阅读理解（共 15 小题；计 30 分）

八、阅读下面短文，根据短文内容选择最佳选项。（共15小题，每小题2分；计30分）

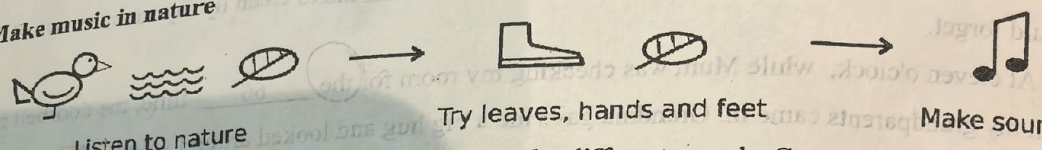
A

Nature Activities Close to Home

Looking for something to do in your spare time? Try these three activities from the National Trust's nature activity list. You can enjoy nature close to home. Read the cards, then use the table to find the activity numbers. The numbers show places on the list, not minutes.

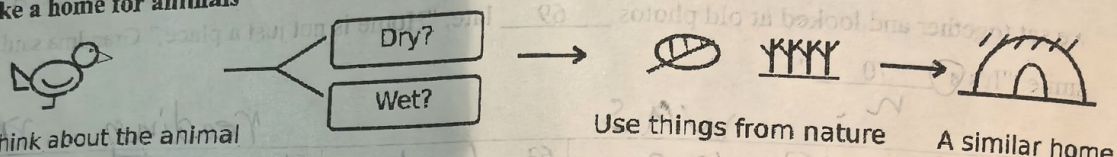
Activity number	Activity name
25	Make music in nature
36	Make a home for animals
44	Watch a bird

Make music in nature



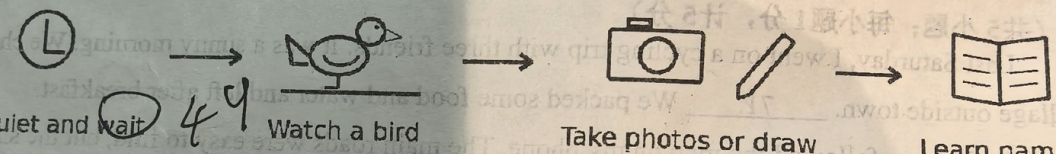
Go outside and listen. Birds, water and trees make different sounds. Can you make a sound like a bird? Your hands and feet can make music too. Walk on dead leaves or hold them in your hands to make sounds. Try different things and listen to their sounds.

Make a home for animals



Think about the animal first. Does it need a dry or wet home? Use things from nature, such as moss and leaves. Make the home similar to the animal's own home. Animals choose where to live. An empty home today may have an animal in it next week.

Watch a bird



Start in your garden or a park near home. Be quiet and wait. Birds may be shy, so loud sounds can make it harder to see them. Take photos or draw pictures to remember the birds. A book about birds can help you learn their names.

46. What does the table tell us about "Make music in nature"?

A. It is one of 25 things.
B. It is No. 25 on the list.
C. It takes about 25 minutes.

47. Tom wants to make sounds with dead leaves. Which number should he look for?

A. 25
B. 36
C. 44

48. Why should you think about the animal first in the second card?

A. To make a home like its own.
B. To let it choose where to live.
C. To know when it will come.

49. What do the last two cards both ask you to do?

A. Use natural things.
B. Keep very quiet.
C. Give it some time.

50. What are the three cards mainly about?

A. Ways to find and draw birds.
B. Ways to have fun in nature around us.
C. Ways to make homes for wild animals.

B

When Yuan Lei first came to our class, nobody wanted to sit with her. She was very quiet. She didn't talk much, and she never put up her hand in class. I'm outgoing, and I love talking, so I thought we were too different to be friends.

Then our science teacher put us in the same group for a project about bees. "Oh no," I thought. On the first day, I did all the talking. Yuan Lei just listened and wrote things down in a small notebook. I was sure she wasn't interested.

I was wrong. The next day, she showed me her notebook. It was full of beautiful drawings of bees and flowers, with notes beside them. "I read a lot about bees," she said softly. "Did you know a bee visits up to 100 flowers on one trip?" I didn't.

We worked together for two weeks. I talked to the class, and she drew the pictures. Our project was the best in the class! "You two are a great team," our teacher said.

Now Yuan Lei is one of my best friends. She is still quieter than me, and I'm still louder than her. But that's OK. She helps me slow down and listen, and I help her ***speak up***. A friend doesn't have to be like you. Sometimes a friend who is different makes you better.

51. How was Yuan Lei different from the writer at first?

A. She was more outgoing.　　B. She talked less in class.

C. She had more friends in class.

52. Why did the writer say "I was wrong" in Paragraph 3?

A. Yuan Lei talked more than the writer.　　B. Yuan Lei made many friends in class.

C. Yuan Lei knew a lot about bees.

53. How did the writer and Yuan Lei share the work?

A. Each did what she was good at.　　B. Yuan Lei did most of the work.

C. The writer drew and Yuan Lei talked.

54. What does "speak up" probably mean in the last paragraph?

A. Talk more.　　B. Speak slowly.　　C. Listen carefully.

55. What does the writer want to tell us?

A. Friends should be like each other.　　B. Friends can learn from differences.

C. Different friends often have problems.

C

Every autumn, something amazing happens in Chengdu. The ginkgo trees along the streets turn bright yellow, and thousands of people come to take photos. But why do green leaves change colour?

Chlorophyll (叶绿素) gives leaves their green colour. In spring and summer, leaves use sunlight to make food from water and air. There is a lot of chlorophyll in the leaves, so the yellow and orange colours are hard to see.

In autumn, the days get shorter and the weather gets colder. There is less and less chlorophyll in the leaves, and the green colour becomes weaker. The yellow and orange colours become easier to see. They were already there in summer.

Red leaves are a little different. Some trees make a new red colour in autumn. Some scientists think it may work like sunglasses and help protect the leaves from strong sunlight. This may help the tree take back useful things from the leaves before they fall.

Pine (松) trees are different. Their leaves can live for more than one year. Some older leaves turn yellow and fall, but many green leaves stay on the tree through winter.

So the next time you walk under the yellow ginkgo trees, remember: those bright colours were there all along, waiting for autumn to show them.

56. What does chlorophyll help leaves do?

A. Take water from the air. B. Make food with sunlight. C. Grow well without light.

57. How are yellow and red leaf colours different in autumn?

A. Yellow is new; red is already there. B. Red can be new; yellow is already there.

C. Both yellow and red are already there.

58. Why does the writer talk about sunglasses?

A. To explain how the red colour helps leaves. B. To show how people enjoy the autumn colours.

C. To explain why trees need more sunlight.

59. Why do pine trees stay green in winter?

A. Their leaves change from yellow back to green. B. Their old leaves fall together before winter.

C. Some green leaves stay on them through winter.

60. What is the best title for the text?

A. A Walk under Chengdu's Ginkgo Trees B. How Leaves Make Food for Trees

C. Why Trees Look Different in Autumn

B 卷（50 分）

一：短文填空。从下面方框中选出 10 个单词，将其正确形式填入短文，使短文意思正确通顺，每词限用一次。（共 10 小题；每题 1 分，计 10 分）

be	big	clear	feel	make	quiet	read	remind	set	three	until	visit

I missed my grandparents so much. Last Saturday, they came ___61___ our new home for the first time since we moved in March. We got up early because there ___62___ a lot to do.

Dad put a new clock on the living-room wall. It is much bigger than the old one, as Grandpa doesn't see very ___63___ these days. I moved a soft chair into their room and placed it by the window. Grandpa loves

___64___. He can do that in the afternoon sun. Mum ___65___ me to clean up my room — twice! She knew I would forget.

At eleven o'clock, while Mum was checking my room for the ___66___ time, the doorbell rang. When my grandparents came in, Grandma gave me a big hug and looked around slowly. "This new home is much ___67___ than the old flat," Grandpa said. "Back at the old flat, cars went past all night!" We were all glad to hear that.

In the evening, Grandma made dinner in our new kitchen, and I helped her ___68___ the table. The dumplings were delicious, especially the ones with cabbage, which she had made just for me. After dinner, we sat together and looked at old photos ___69___ late. "Home is not just a place," Grandma said with a smile. "It's a ___70___."

61. to visit	62. were	63. dearly	64. ~~reminded~~	65. ~~reminded~~
66. third	67. ~~quiet~~	68. to set	69. quiet	70. reminder

二、补全短文。根据短文内容，从短文后的A---F选项中，选出5个适当的选项补全短文。

（共5小题；每小题1分，计5分）

Last Saturday, I went on a cycling trip with three friends. It was a sunny morning. We chose a small village outside town. ___71.___ We packed some food and water and left after breakfast.

At first, we followed the map on my phone. The main roads were easy to find, but the smaller ones were not. At a crossing, we stopped. ___72.___ A driver passing by seemed to find our trouble. He stopped and showed us the right road, and we reached the village an hour later.

After lunch, we walked around in the village and saw many interesting plants. We knew some of them, but others were new to us. ___73.___ We looked at the pictures on its screen. Soon we could name a few more plants. We took photos of our favourite ones.

On the way home, Sandy's bike broke down. Daniel knew how to fix it, but it was not a job for one person. ___74.___ Some held the bike while others helped Daniel. Sandy tried it again, and we could finally ride on.

That evening, I looked at our photos. They showed the amazing scenery along the way, the village and the plants. ___75.___ I also remembered the kind driver and our friends working together. The trip was not always easy, but it gave us something special to remember and share.

A. We chose to find out more after the trip.
B. No one knew which way to go from there.
C. We wanted to get close to nature for a day.
D. My phone helped us learn more about the plants.
E. But they could not show every important part of the trip.
F. We all worked together to get the bike moving again.

71. C	72. B	73. D	74. F	75. E

三、阅读表达（共10 小题，计20 分）

A. 完成图表。根据短文内容，完成图表中所缺信息。每空不超过3 个单词。（共5 小题；每小题2 分，计10 分）

On the coast of Andenes in northern Norway (挪威北部), people are building a new museum. Its name is The Whale. Why build it here? Whales come close to this coast whether it's in spring, summer or in autumn and winter, so Andenes is one of the best places in the world to see them. The museum opens on June 4th, 2027.

A designer (设计师) from Denmark planned the building. Its round roof looks like the back of a whale coming out of the sea. Inside, there is nothing in the middle to hold up the roof, so the hall feels wide and open to the sea and sky.

Visitors can walk up onto the roof and look out over the Norwegian Sea. For weeks in summer, the sun stays up all night, so the sky is still bright at midnight. On dark winter nights, lucky visitors can see the Northern Lights (北极光) dancing across the sky.

Inside, visitors can meet five huge wooden killer whales (虎鲸). Each one is the same size as a real killer whale. In fact, they are copies (复制品) of five killer whales living near Norway. In another room, real whale bones help visitors learn about a whale's body. There is also a dark sound room. Visitors can sit there quietly and listen to whales "talking" to each other.

One part of the museum tells the story of people and whales. Long ago, people caught a great number of whales for meat and oil, and some kinds almost disappeared. Today, more and more people know whales are clever and amazing animals, and they want to protect them. After their visit, people can also take a boat trip to see real whales. The Whale hopes to help people understand whales better and care about them more.

THE WHALE

A Visit from Land to Sea

Before you go: basic information

Andenes, northern Norway. Planned opening: 4 June 2027.

Whales come close to the coast ____76.____.

The building and roof

A whale-shaped roof covers a big and wide hall whose middle area is ____77.____.

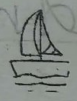
Sea views; bright summer nights; winter lights.

Inside the museum

Wooden whales: Their size is ___78.___ that of the real animals they copy.

Bones: learn about bodies. Sound room: hear whale voices.

People and whales



People get ___79.___ of whales.

Past: meat and oil. Today: whales are clever, should be protected

Afterwards: meet real whales on ___80.___.

76. every seasons

77. empty

78. the assame as

79. ramindiny

80. a boort trip

better understanding

B. 任务型阅读。根据短文内容，按要求回答问题。（共5 小题；每题2 分，计10 分）

LETTERS ACROSS THE SEA

Last weekend, my grandma asked me to see the movie *Dear You* with her. To be honest, I didn't want to go because I thought a movie about old letters was boring.

But the movie surprised me. Long ago, many people from Chaoshan, Guangdong, left their hometowns to work in countries like Thailand. They couldn't go home easily, so they sent letters with money inside to their families. People call these letters Qiaopi (侨批), or "silver letters".

In the movie, a young man goes to Thailand to look for his grandfather, but there he learns some sad news: his grandfather died many years ago. Then who sent all those letters and money to his grandma? It was a kind woman, Xie Nanzhi. She didn't want his grandma to feel sad, so she pretended (假装) to be his grandfather and wrote to her for 18 years.

On the way home, Grandma was quiet. Then she took out an old letter from her bag and said, "My father sent this from Thailand." I held it carefully and saw the paper was yellow, but the words were still clear. At that moment, I felt something special.

Now I know old letters are not boring at all. From them, I can see our ancestors (祖先) were brave, hard-working and honest. These letters also connect me with my family's past and my hometown.

Comments（评价）

Lucy: I think Xie Nanzhi's lie was beautiful because it came from love and gave the whole family hope for 18 years.

Mike: Sorry, Lucy, but I don't agree. A lie is still a lie, even a kind one, and being honest is always better.

Sophia: Old things can tell us so much about the past. Last night I asked my grandpa about his past, and he told me many interesting stories.

Tom: Today we send messages on our phones, but we forget them quickly. A letter on paper is warmer and lasts longer, so old letters are special.

81. Why did many people from Chaoshan send Qiaopi to their families long ago?

Because they (old couldn't go h

82. Why did Grandma take out the old letter on the way back home?

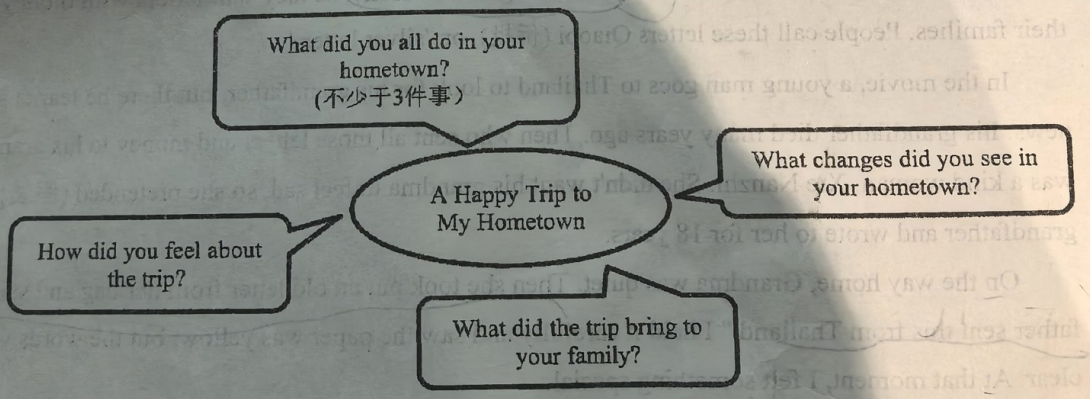
83. How did the writer's feelings about old letters change?

84. Which two people agree with the writer about old letters?

85. What do you think of Qiaopi? (Reply in your own words)

四、书面表达。（计 15 分）

在刚刚过去的国庆假期，你和爸爸妈妈开车回老家看望爷爷奶奶，请你以 A Happy Trip to My Hometown为题写一篇80词左右的英语短文记下你的所见、所闻、所做和所感。



要求：1. 合理想象，有细节；

2. 注意信息整合，条理清楚；

3. 注意使用适当的时态和句型；

4. 书面整洁。

嘉祥教育集团 2026-2027 学年初二上学期质量监测试题

物　理

注意事项：

1. 全卷分为第 A 卷和第 B 卷，A 卷满分 100 分，B 卷满分 20 分；考试时间为 90 分钟。
2. 考生使用答题卡作答。
3. 在作答前，考生务必将自己的姓名、考号涂写在试卷和答题卡规定的地方。考试结束，监考人员只将答题卡收回，试卷请考生自己妥善保存。
4. 选择题部分必须用 2B 铅笔填涂；非选择题部分必须使用 0.5 毫米黑色墨水签字笔书写，字体工整、笔迹清楚。
5. 请按照题号在答题卡上各题目对应的答题区域内作答，超出答题区域书写的答案无效；在草稿纸、试卷上答题均无效。
6. 保持答题卡清洁，不得折叠、污染、破损等。

A 卷(共 100 分)

一、选择题(本大题共 15 个小题，每小题 2 分，共 30 分。在每小题给出的四个选项中，只有一项符合题目要求)

1. 下列估测数据中，不合理的是（B）
   A. 嘉祥 160A 运动外套衣长约为 65cm　　B. 初中生百米赛跑平均速度约为 7km/h
   C. 完整播放一遍我国国歌用时约为 46s　　D. 中学生使用的课桌的高度约为 80cm
2. 练习使用刻度尺测量物理课本宽度时，下列说法正确的是（B）
   A. 刻度尺未与课本边平行，测量结果不准，属于测量误差
   B. 受仪器和测量方法限制，测量过程中总会产生误差
   C. 实验过程中所选取刻度尺的测量范围越大越好
   D. 测量时为了减小误差可以尝试多估读几位
3. 小嘉用毫米刻度尺测量物理课本的长度，五次读数分别为 26.13 cm、26.12 cm、26.10 cm、26.11 cm、26.31 cm。该物理课本的长度应为（C）
   A. 26.115cm　　B. 26.154 cm　　C. 26.12 cm　　D. 26.15 cm
4. 下列描述的内容中，属于机械运动的是（B）
   A. 电子绕原子核运动　　B. 秋天树叶纷纷飘落
   C. 植物生长　　D. 五四运动
5. 某物体在做匀速直线运动，关于速度公式 $v=\frac{s}{t}$，下列说法正确的是（D）
   A. 物体通过的路程越长，速度越大
   B. 物体运动的时间越短，速度越大
   C. 物体运动的速度与路程成正比，与时间成反比
   D. 物体做匀速直线运动通过的路程与时间成正比
6. 如图 1 所示，*a*、*b* 两小车同时同地向东运动，甲是 a 车的 *s-t* 图像，乙是 b 车的 *v-t* 图像，下列判断正确的是（C）
   A. 0-2s，*a* 车做加速运动
   B. 0-6s，*b* 车处于静止状态
   C. 0-6s，*a* 车平均速度大于 *b* 车速度
   D. 以 *a* 车为参照物，*b* 车一直向西运动

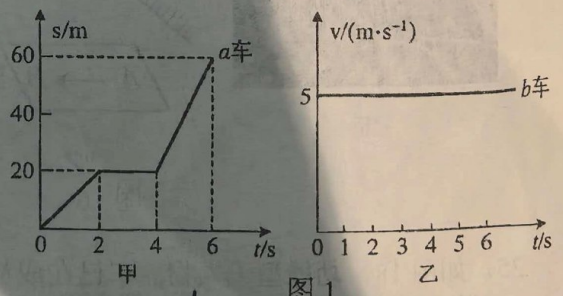


图 1

7. “三更灯火五更鸡”出自唐代颜真卿勉励学子勤学的诗句，诗句提及古代夜间重要报时方式——打更，旧时打更人打梆子（竹筒）或敲铜锣巡夜报时。下列有关打更的说法正确的是（A）
   A. 打更声是以波的形式传播　　B. 停止打更后，更声会立即消失
   C. 打更声可以在真空中传播　　D. 打更声是由空气的振动产生的

8. 如图 2，中国科学院院士褚君浩在北京现场展示了利用特殊材料实现的“隐身术”。据了解，该“隐身术”借助了能让光线产生规律折射的神奇材料——“柱镜光栅”。下列现象与该“隐身术”的光学原理相同的是（ A ）

A. 潭清疑水浅　　B. 水中捞月

C. 对影成三人　　D. 一叶障目

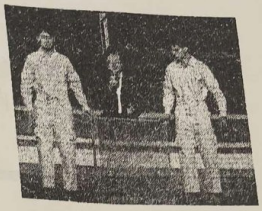
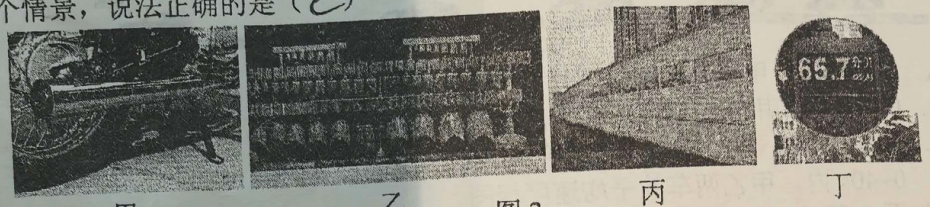


图 2

9. 控制噪声已成为城市环境保护的重要项目之一。下列关于图 3 中的几个情景，说法正确的是（ C ）



甲　　乙　　丙　　丁

图 3

A. 图甲：摩托车的消声器是在接收处控制噪声的

B. 图乙：敲击编钟发出的美妙声音不会成为噪声

C. 图丙：路旁的隔声板是在传播过程中控制噪声

D. 图丁：路旁的噪声监测仪可以测量和减弱噪声

10. 乐器主要分为三种类型：打击乐器、弦乐器和管乐器。关于图 4 中的乐器，下列说法正确的是（ C ）

腰鼓　　古筝　　排箫

图 4

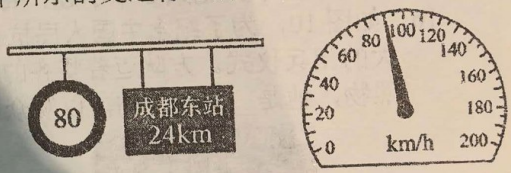
A. 拍击腰鼓时，鼓面振动越快，响度越大

B. 弹奏古筝时，琴弦调得越紧，音调越低

C. 合奏同一首乐曲时，三种乐器发出声音的音色一般不同

D. 合奏同一首乐曲时，三种乐器发出声音的响度一定相同

11. 秋假期间，小嘉乘车去成都东站，他看到了图 5 甲所示的交通标志牌和图 5 乙所示的汽车仪表盘。下列说法正确的是（ B ）



甲　　图 5　　乙

A. “80”的含义是最高车速为 80m/s

B. 该仪表盘显示该车的速度为 90km/h

C. 根据仪表盘可知这辆汽车没有超速行驶

D. 从标志牌到成都东站不超速最快需 30min

12. 如图 6 所示，小嘉利用激光笔、可向后翻折的硬纸板、平面镜等器材探究光的反射定律。下列说法正确的是（ C ）

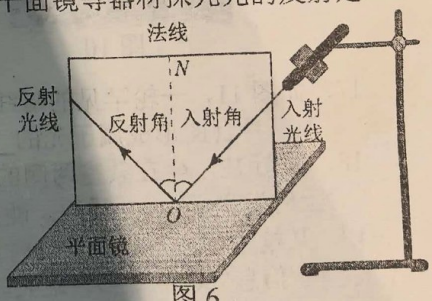


图 6

A. 为了便于观察光路，应尽量在亮的地方进行实验

B. 在选择硬纸板作为光屏时应尽量选择光滑一些的

C. 硬纸板向后翻折看不到反射光，但反射光仍存在

D. 多次改变入射角大小进行实验，是为了减小误差

13. 成都白鹭湾湿地公园里，一只白鹤静立在平静的水中，在岸边可以看到白鹤的两个“影”——影子和倒影，如图 7。下列分析正确的是（ A ）

图 7

A. 白鹤的影子是由光的直线传播形成的

B. 白鹤水中的倒影是由光的折射形成的

C. 白鹤飞离水面时，它的倒影位置不变

D. 白鹤站立在水中时，它的倒影是实像

14. “频闪摄影”能够记录物体在相等时间间隔所处的位置。甲、乙两车同时同起点在同一段平直公路同向行驶，图 8 为两车运动的频闪照片。下列说法正确的是（ D ）

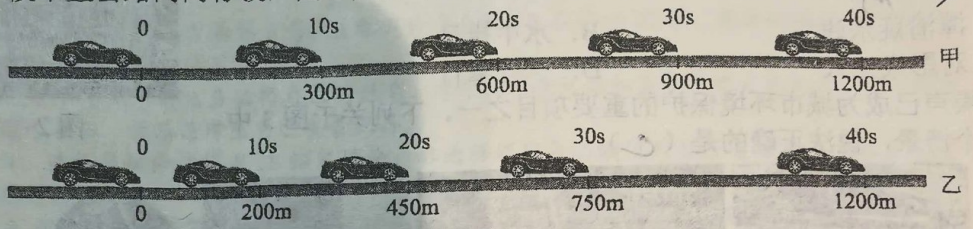


图 8

A．0~40s 内，甲车一定做匀速直线运动
B．0~30s 内，甲车的速度一直比乙车快
C．20s~30s 内，甲、乙两车不可能相遇
D．0~40s 内，甲乙两车的平均速度相同

15. 如图 9 所示为初中物理教材部分实验插图，关于各图说法不正确的是（ D ）

甲　　乙　　丙　　丁

图 9

A．甲图：探究降落伞滞空时间影响因素，用到了控制变量法
B．乙图：探究音调高低与振动快慢关系时，需控制响度相同
C．丙图：若平面镜与半圆形光屏不垂直，光屏上将看不到反射光线
D．丁图：光从空气斜射向水中，增大入射角，折射光线可能会消失

二、填空题(本大题共 8 个小题，每空 2 分，共 32 分)

16. 如图 10，为了纪念中国人民抗日战争暨世界反法西斯战争胜利 80 周年，我国进行了盛大的阅兵仪式。方队迈着整齐的步伐通过天安门广场，队中的某战士以他身边战士为参照物，他是_____的，以观众席上的观众为参照物，他是_____的。

图 10

图 11

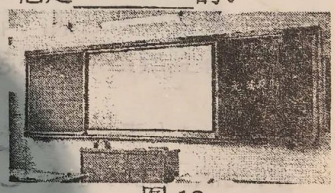


图 12

17. 如图 11，一轮罕见的暗红色“红月亮”悬挂夜空。月亮_____（选填“是”或“不是”）光源。月食的形成属于光的_____（选填“直线传播”、“反射”或“折射”）。

18. 如图 12，坐在教室两侧的同学有时候看不清多媒体屏幕上的字，这是光在光滑的屏幕上时发生了_____反射；能在不同方向看清黑板上的粉笔字是因为光发生了_____反射。

19. 某校进行地震逃生演练时，听到警报，同学们有序疏散，这是利用声音传递_____（选填“信息”或“能量”）。地震时产生的_____（选填“超声波”或“次声波”）可用于监测预警。

20. 如图 13 左图，博物馆人员为排查某个带有隐形裂痕的编钟，将甲、乙、丙、丁四个编钟由大到小依次排列。用相同木槌分别敲击四个编钟，产生的声波输入示波器，得到图 13 右图的四组波形。可推测：波形 4 是编钟_____（选填“甲”、“乙”、“丙”、“丁”）产生的声波；波形_____（选填“1”、“2”、“3”、“4”）对应的编钟最可能存在隐形裂痕。

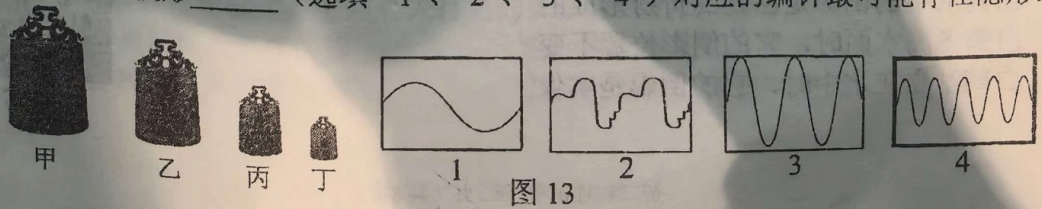


图 13

21. 国内首个六人制棍网球球场落地嘉祥。校队队员甲、乙在球场匀速跑动，运动相同路程，甲、乙所需时间之比为 3:5，已知甲的速度为 3m/s，则乙的速度为____m/s；若一开始乙在甲前方 12m 处，两人沿同一直线同向运动，同时出发后至少经过____s，甲、乙相距 4.8m（场地足够大）。

22. 如图 14 所示，一块平面镜直立在水平地面上，小嘉沿与镜面平行的直线从 A 走到 D，从平面镜底边两端点向 AD 直线所作垂线的交点分别为 B、C。小嘉在 AB 段或 CD 段行走时，平面镜中_____（选填“能”或“不能”）形成自己的像；若小嘉沿 BC 段行走的速度为 1m/s，则以小嘉为参照物，镜中的像速度为_____m/s。

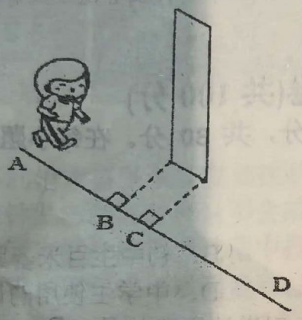
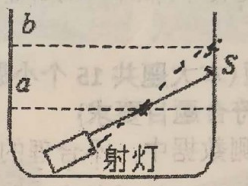


图 14



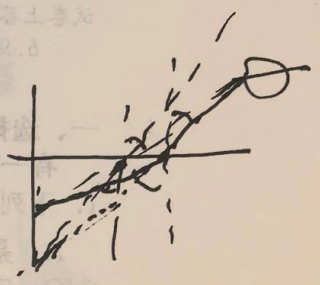


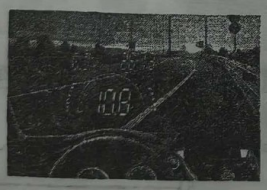
图 15

23. 如图 15 所示，在空水槽底部装了一个射灯，射灯发出一束光在右侧壁上形成一个光斑，在光斑处固定一小块橡皮泥 S。向水槽内缓慢加水，当水面由 a 位置上升至橡皮泥 S 所在高度的过程中，光斑的位置_____（选填“向上移动”、“向下移动”或“静止不动”）；当水面升至 b 位置时，眼睛在水面左上方透过水面观察此时的光斑和橡皮泥，会看到光斑静止在橡皮泥的_____（选填“下方”、“左上方”“右上方”或“同一位置”）。

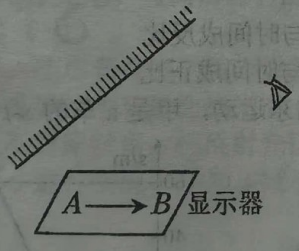
三、作图与计算题（计算题在解答时应写出公式和重要的演算步骤，只写出最后答案不能得分。24 题每小问 2 分，25 题和 26 题每题 6 分，共 16 分。）

24.（1）如图 16 甲所示为汽车安装的抬头显示系统（简称 *HUD*），*HUD* 可将水平显示器上重要的行车数据通过前挡风玻璃（看作平面镜）呈现在司机正前方，方便安全行车。请你在图乙中利用平面镜成像规律，作出显示器内箭头 AB 的像 $A'B'$。（保留作图痕迹）

（2）如图 17 甲是小嘉观察水缸中鱼的场景，从侧面观察，光经缸壁折射进入眼睛。图 17 乙中所示的 S 位置代表鱼的实际位置，S' 位置代表看到的鱼的虚像位置。请在图 17 乙中画出小嘉看到鱼的光路图。



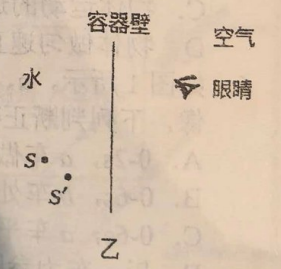
甲



乙

图 16

甲



乙

图 17

25. 如图 18，功能型无人物流车已在成都多地投入运营。某次无人物流车从快递网点出发，前往相距 1.8 km 的社区驿站：前 20s 行驶了 100m；接着以 36km/h 的速度匀速行驶了 1300m；之后在路口等待 20s，再用时 30s 抵达驿站并停车。求：

（1）无人物流车以 36km/h 的速度匀速行驶 1300 m 所用的时间。

（2）无人物流车全程的平均速度。

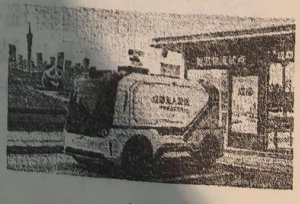


图 18

26. 2026 年 10 月 10 日，“蛟龙”号载人潜水器抵达西太平洋东加罗林海盆作业区开展深海科考作业。如图 19 为“蛟龙”号在海底探索时的情景。“蛟龙”号可使用超声波在海底定位，在一次任务中，静止的“蛟龙”号向海底发射第一束超声波，经过 6s 后收到回波信号，随后“蛟龙”号立即以 5m/s 的速度匀速下潜至海底，在下潜途中向海底发射了第二束超声波，4s 后收到回波信号。已知声音在海水中传播速度为 1500m/s。求：

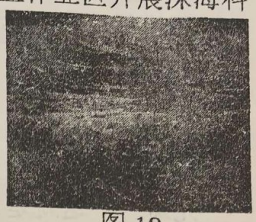


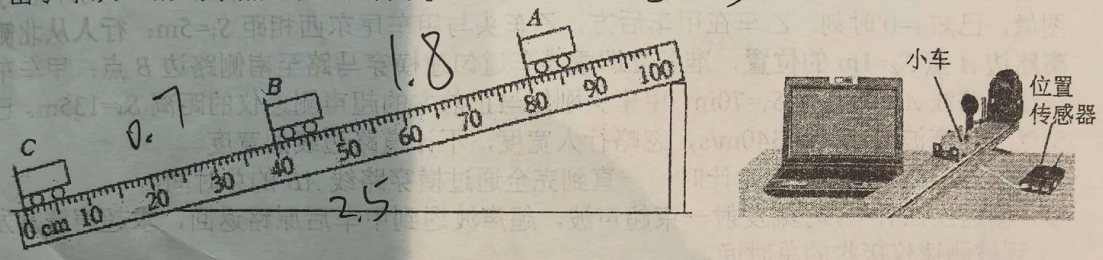
图 19

（1）“蛟龙”号接收到第一束回波信号后，下潜多长时间才能到海底？

（2）发射第二束超声波时，“蛟龙”号距离海底的距离为多少？

四、实验探究题(共 2 小题，每空 2 分，共 22 分)

27. 如图 20 甲，小嘉利用斜面装置测量小车沿斜面下滑的平均速度。他将小车从坡顶 A 处静止释放，测出小车从 A 滑到坡底 C 处的时间 $t_1=2.5\text{s}$；再次将小车从 A 处静止释放，测出小车从 A 滑到中点 B 处的时间 $t_2=1.8\text{s}$。



甲　　图 20　　乙

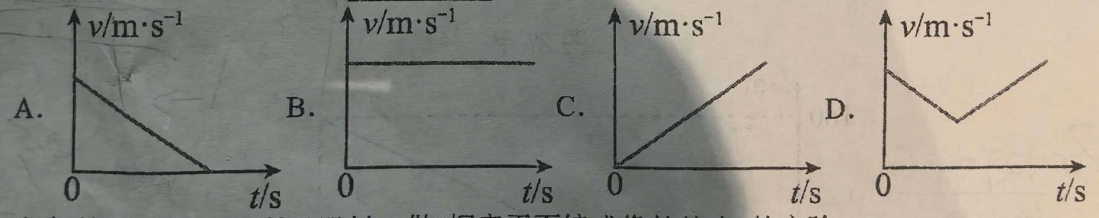
（1）本实验测平均速度的原理是_____；

（2）由图甲可知，小车从 A 点到 C 点的路程为_____m；

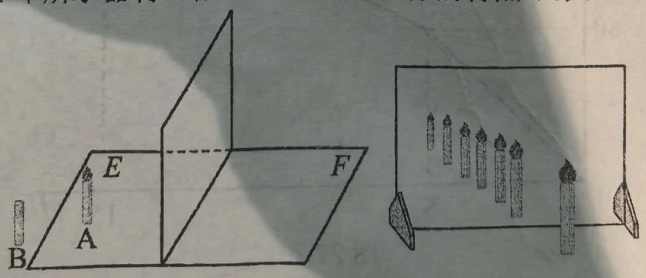
（3）组装器材时，用木块垫高斜面一端。若斜面坡度过大，小车下滑太快，难以及时停止计时，从而导致测出的平均速度_____（选填“偏大”、“偏小”或“无影响”）；

（4）由小嘉测量的数据可判断：小车在前半程的平均速度_____全程的平均速度；（选填“大于”、“等于”或“小于”）

（5）为精准研究小车的运动状态，小嘉采用图 20 乙装置开展实验：位置传感器借助超声波采集不同时刻小车的位置信息，计算机处理实时数据，输出小车对应时刻的速度。下列图像符合小车运动规律的是_____。



28. 小嘉利用如图 21 甲所示器材，做“探究平面镜成像的特点”的实验。



甲　　图 21　　乙

（1）实验室有厚度为 1 mm、5 mm 两种玻璃板，应选用_____（选填“1”或“5”）mm 厚的玻璃板进行实验。

（2）小嘉拿另一支完全相同的蜡烛 B 在玻璃板后面移动，人眼应在_____（选填“E”或

“F”）一侧观察，直到看上去蜡烛 B 跟蜡烛 A 的像完全重合。

（3）小嘉借来了旁边实验小组的玻璃板，放在蜡烛 A 的后面，他惊奇地发现，蜡烛 A 成了很多个像，如图 21 乙所示。这些像与蜡烛 A 的大小关系是_____（选填“等大”、“越远越缩小”或“越远越放大”）。

（4）小嘉用一张黑纸将玻璃板上半部分全部遮住，通过下半部分，平面镜可以成_____；

A．蜡烛 A 下半部分的像　　B．蜡烛 A 上半部分的像

C．蜡烛 A 完整的像　　D．比蜡烛 A 更小的像

（5）保持其它条件不变，若小嘉换用一个面积更大的玻璃板来做实验，则_____。

A．蜡烛 A 的像会变大　　B．蜡烛 A 的像会向上移动

C．能观察到蜡烛 A 的像的范围会变大　　D．蜡烛 A 的像到玻璃板的距离会变大

（6）小嘉和同学为了进一步探究像与物是否左右相反，选用 LED 光源进行实验，现有“A、F、M、T、V、X”六种字母形状的光源可供选择，小嘉的最佳选择是_____（选填字母）。

B 卷　（共 20 分）

一、不定项选择题（每小题 2 分，共 10 分，每小题有的一个选项符合题目要求，有的多个选项符合题目要求，对而不全 1 分，选错 0 分）

1. 在嘉祥组织的各类活动和素质课程中，同学们开展了多项长度测量实验，下列说法不正确的是（ B ）

A. 嘉祥 AI 创意实践大赛中，小嘉多次测量智能小车长度，取平均值来减小误差

B. 嘉祥物理实验选修课上，刻度尺零刻度线磨损，则该刻度尺无法继续用来测量

C. 嘉祥杯比赛前，用刻度尺测量 APM 机器人机械臂长度时，视线需与尺面垂直

D. 嘉祥科技节活动中，测量薄塑料片厚度，可用累积法测总厚度再除以薄片张数

2. 在 2026 年嘉祥杯青少年棍网球锦标赛中（图 22），两校队展开激烈角逐。小嘉带球向对方球门奔跑时，发现队友甲从正前方朝自己跑来接应，同时又发现对方球员乙正迅速从正后方朝自己快速逼近，眼看快要被追上，于是小嘉迅速将球抛出传给前方队友甲。关于该情景，下列说法正确的是（ C ）

A. 三名球员中，球员乙的速度最快

B. 以队友甲为参照物，球员乙可能静止

C. 以对方球员乙为参照物，小嘉在向后运动

D. 以正在抛出的棍网球为参照物，队友甲可能静止

图 22

3. 嘉祥八年级举行了创新实验大赛，比赛中，同学们展示了图 23 所示甲、乙、丙、丁四组声现象创新实验，下列说法正确的是（ D ）

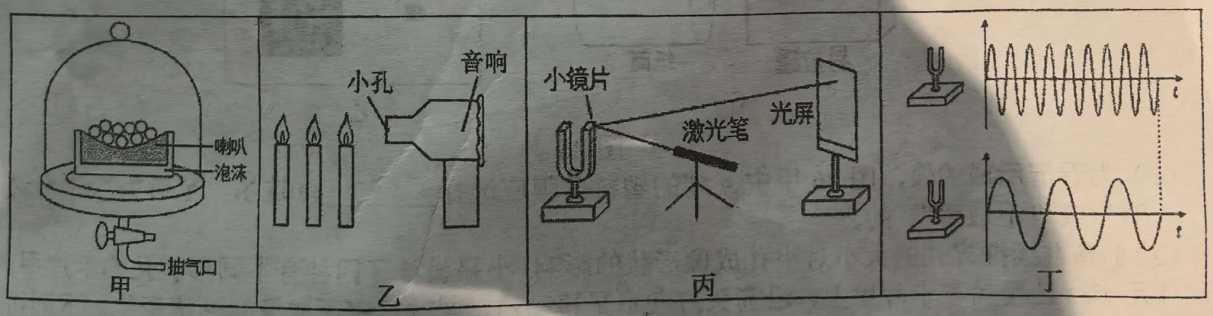


图 23

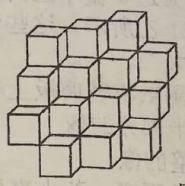
A．图甲：抽出罩内空气，喇叭上的小球跳动幅度不变，说明声音的传播不一定需要介质

B．图乙：音响发声，离音响越近的蜡烛，晃动越明显，说明音调与到声源的距离有关

C．图丙：敲击音叉，通过光屏上光斑来回摆动，将音叉的微小振动放大，便于观察

D．图丁：把两个振幅相同的音叉靠近，敲击其中一个音叉，另一个能够共鸣发声

4. “鹊桥”是嫦娥四号月球探测器的中继卫星，是中国首颗、也是世界首颗地球轨道之外专用中继通信卫星。“鹊桥”上安装了激光角反射器，开展了超地月距离的反射式激光测距试验。如图 24 甲所示，角反射器由相互垂直的小平面镜组合而成，图 24 乙为两块平面镜组合的反射器简图，当地面激光站发出激光射向角反射器，激光能被反向射回地面，已知激光往返总共用时约 3s。下列说法正确的是 B D


甲

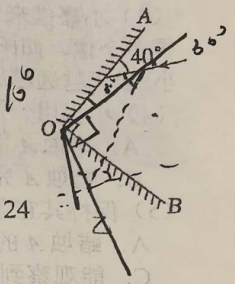


图 24

A. 地月测距通常选用激光而非超声波，因为超声波传播速度过小

B. 若光速取 3×10^8m/s，则地面激光站到“鹊桥”距离约 4.5×10^8m

C. 将图乙中的反射器绕 O 点旋转 10°，最终出射光线将旋转 20°

D. 将物体置于图乙反射器的正右方，通过反射器可以成 3 个虚像

5. 嘉祥研学徒步活动中，A、B 两个班级队伍沿平直道路同向匀速向前行进，B 班队伍在 A 班前方。A 班一名同学从 A 班队伍出发，跑步给前方的 B 班队伍送物品，物品送达后立刻原路折返回到 A 班，该同学往返跑步的速度大小不变，不计队伍长度。图 25 为 B 班队伍、A 班队伍、送物品同学运动的 S-t 图像。已知 B 班前进速度 v_B=0.8m/s，送物品同学的速度为 5m/s。针对这个过程，下列说法正确的是（ C D

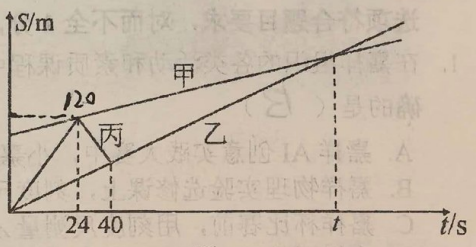


图 25

A. 甲图像代表 A 班队伍的运动情况

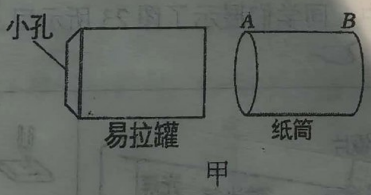
B. A 班和 B 班队伍最初相距 120m

C. A 班队伍前进的速度大小为 1m/s

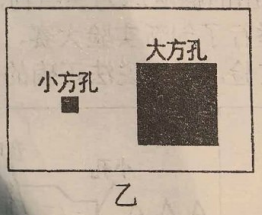
D. A 班追上 B 班队伍需要用时 504s

二、综合题（本大题共 2 小题，6 题 4 分，7 题 6 分，第 7 题在解答时应写出公式和重要的演算步骤，只写出最后答案的不能得分，共 10 分。）

6. 为响应“做中学”理念，小嘉用易拉罐自制了一个简易针孔相机，如图 26 甲所示。他将易拉罐盖子去除，在底部中心钻个小孔，AB 为两端开口的圆形纸筒，用塑料薄膜将纸筒 A 端蒙住，用于承接所成像，然后将纸筒 A 端插入易拉罐，做小孔成像实验。



甲

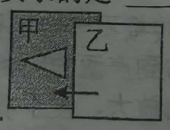


乙

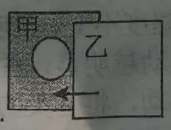
图 26

（1）为看清所成的像，图 26 甲中 A 端的塑料薄膜应选择_____（选填“全透明”、“半透明”或“不透明”）的。

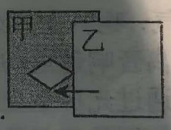
（2）小嘉继续探究孔的大小对小孔成像形状的影响，小嘉设计了四种有不同形状孔的卡片甲，用卡片乙覆盖在卡片甲上，沿箭头方向水平移动乙，同时观察所成像的形状变化。下列符合要求的是 _____ 。



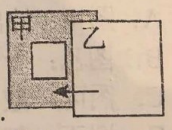
A.



B.



C.



D.

（3）小嘉取一块硬纸板，上面挖有两个孔（小方孔边长约 2 mm，大方孔边长约 4cm），如图

26 乙所示。他将硬纸板正对太阳，将硬纸板调整到距离地面合适的高度，对地面形成的光斑进行观察，光斑的大小、形状可能符合实际的是_____。

A. B. C. D.

（4）为了估测太阳直径，小嘉观察到：正午树荫下，阳光穿过树叶缝隙，在地面形成了圆形光斑——这就是太阳经小孔成的实像。他测出了某一圆形光斑的直径为 1.1cm，光斑到小孔的距离为 1.2m，查阅资料得知太阳到地球的距离为 1.5×10^{11}m，根据小孔成像规律可以估算太阳的直径为_____m。

7. 如图 27 甲所示，一条笔直单向双车道公路，每条车道宽 D=3m。甲车（轿车）和乙车（小货车）分别在各自车道中心线自西向东行驶；甲车车长 $l_{甲}$=5m、车宽 $d_{甲}$=2m，以 $v_{甲}$=36km/h 的速度匀速行驶；乙车车长 $l_{乙}$=8m、车宽 $d_{乙}$=2m，图 27 乙为乙车从 t=0 时刻开始的 s-t 图像。已知 t=0 时刻：乙车在甲车后方，乙车头与甲车尾东西相距 S_1=5m；**行人从北侧距离路边 A 点 S_2=1m 的位置**，准备立即垂直车道匀速横穿马路至南侧路边 B 点；甲车车头与横穿路线 AB 的距离 S_3=70m；甲车头刚好与正前方的超声测速仪的距离 S_4=135m。已知空气中超声波速度 $v_{声}$=340m/s，忽略行人宽度，不计道路边线的宽度。

（1）求乙车从 t=0 时刻开始计时，一直到完全通过横穿路线 AB 的总时间。

（2）测速仪在 t_1=3s 时刻发射一束超声波，超声波遇到甲车后原路返回，求超声波从发出到被测速仪接收的总时间。

（3）为保证行人横穿马路全程安全，求行人匀速横穿马路的安全速度范围。

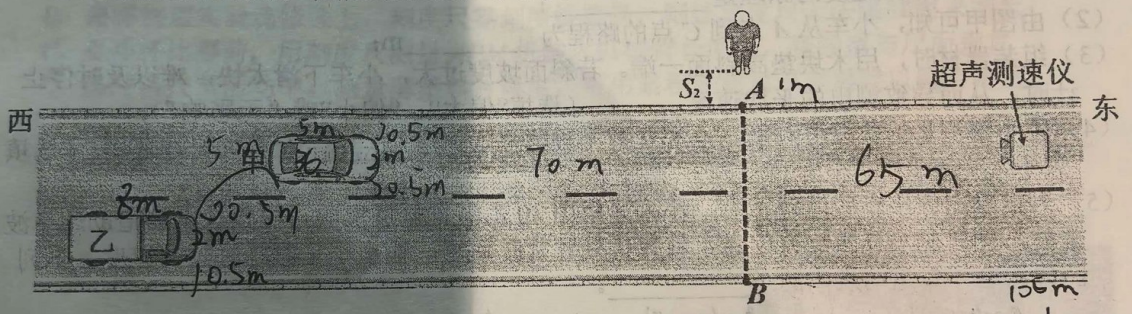


图 27 甲

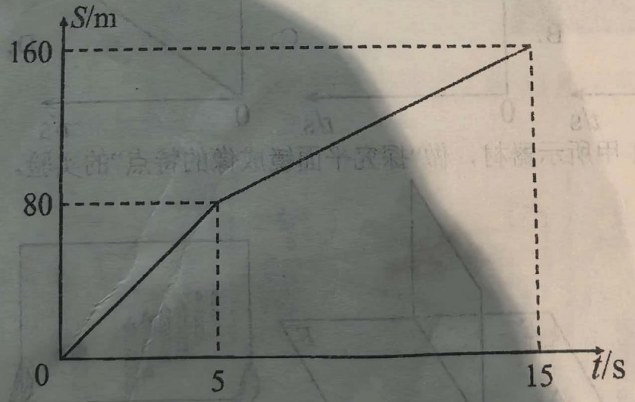


图 27 乙

嘉祥教育集团 2026-2027 学年初二上学期质量监测试题

数 学

注意事项：

1. 全卷分为第 A 卷和第 B 卷，A 卷满分 100 分，B 卷满分 50 分；考试时间为 120 分钟。
2. 考生使用答题卡作答。
3. 在作答前，考生务必将自己的姓名、考号涂写在试卷和答题卡规定的地方。考试结束，监考人员只将答题卡收回，试卷请考生自己妥善保存。
4. 选择题部分必须用 2B 铅笔填涂；非选择题部分必须使用 0.5 毫米黑色墨水签字笔书写，字体工整、笔迹清楚。
5. 请按照题号在答题卡上各题目对应的答题区域内作答，超出答题区域书写的答案无效；在草稿纸、试卷上答题均无效。
6. 保持答题卡清洁，不得折叠、污染、破损等。

A 卷（共 100 分）

第 I 卷（选择题 共 32 分）

一、选择题（本大题共 8 个小题，每小题 4 分，共 32 分．在每小题给出的四个选项中，只有一项符合题目要求）

1. 下列实数是无理数的是（ ）

A. $\frac{22}{7}$　　B. 0.1010010001　　C. $\sqrt{16}$　　D. π

2. 满足下列条件的$\triangle ABC$，不是直角三角形的是（ ）

A. $b^2-c^2=a^2$　　B. $a:b:c=3:4:5$

C. $\angle C=\angle A-\angle B$　　D. $\angle A:\angle B:\angle C=9:12:15$

3. 若 a，b 为实数，其中 a 是 1 的算术平方根，b 是 64 的立方根，则点 $P(a, b)$ 关于 y 轴的对称点 Q 的坐标为（ ）

A. $(1, 4)$　　B. $(-1, 4)$　　C. $(1, -4)$　　D. $(-1, -4)$

4. 若点 P 在第二象限，且点 P 到 x 轴的距离为 4，到 y 轴的距离为 2，则点 P 的坐标为（ ）

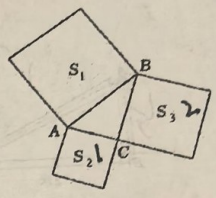
A. $(2, -4)$　　B. $(4, -2)$　　C. $(-4, 2)$　　D. $(-2, 4)$

5. 关于 x，y 的二元一次方程组的解 $\begin{cases}6x+5y=3k,\\2x+3y=2k-1\end{cases}$ 满足 $x+y=3$，则 k 的值为（ ）

A. 5　　B. 6　　C. 7　　D. 8

17-2 18-3

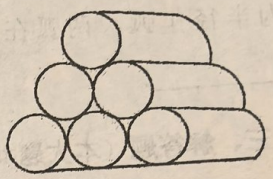
6. 如图，分别以 Rt△ABC 三边为边向外作三个正方形，其面积分别用 S_1，S_2，S_3 表示，已知 $S_2=1$，$S_3=2$，则 S_1 的值为（ D ）



A．1　　B．$\sqrt{2}$　　C．$\sqrt{3}$　　D．3

7. 瓶子或者罐头盒等圆柱形的物体常常如图所示那样堆放着，随着层数的增加，物体总数也会发生变化，数据如下表，则下列说法错误的是（ C ）

层数 n / 层	1	2	3	4	5	…
物体总数 y / 个	1	3	6	10	15	…



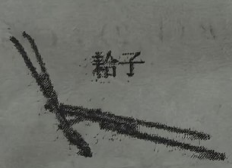
A．在这个变化过程中，层数是自变量

B．当堆放层数为 7 层时，物体总数为 28 个

C．物体的总数 y 随着层数 n 的增加而均匀增加

D．物体的总数 y 与层数 n 之间的关系式为 $y=\dfrac{n(n+1)}{2}$

8. 古代劳动人民在实际生活中有这样一个问题：“耡子耧六十三，百根腿地里钻，两者各几何？”其大意为：耡子和耧共有 63 个，共有 100 条腿，问有多少个耡子，多少个耧？（耡子有一条腿，耧有两条腿）设耡子有 x 个，耧有 y 个，则可列出方程组（ B ）



A．$\begin{cases}x+y=100\\x+2y=63\end{cases}$　　B．$\begin{cases}x+y=100\\2x+y=63\end{cases}$　　C．$\begin{cases}x+y=63\\x+2y=100\end{cases}$　　D．$\begin{cases}x+y=63\\2x+y=100\end{cases}$

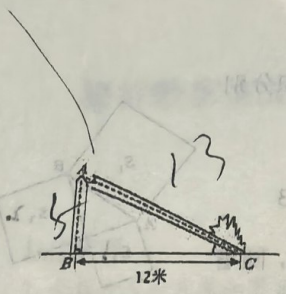
第Ⅱ卷（非选择题　共 68 分）

二、填空题（本大题共 5 个小题，每小题 4 分，共 20 分．答案写在答题卡上）

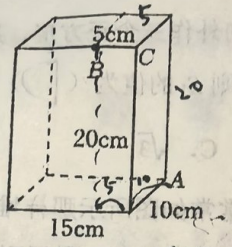
9. 已知 $|a-3|+\sqrt{3+b}=0$，则 $(\dfrac{a}{b})^{2026}=$_____．　-3, -2

10. 将点 $Q(m+2,\ m+3)$ 向左平移 3 个单位长度，向上平移 2 个单位长度得到点 P，点 P 恰好落在 x 轴上，则点 P 的坐标是_____．　m-1, m+5 <

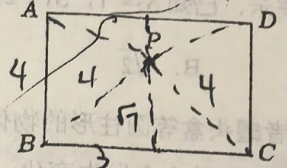
11. 如图所示，一棵树被风刮断了，树顶落在离树根 12 m 处，折断处的高度 AB 为 5 m，则这棵树折断前高_____m．



第 11 题图



第 12 题图



第 13 题图

12．如图，一个长方体形盒子的长为 15 cm，宽为 10 cm，高为 20 cm，点 B 到点 C 的距离是 5 cm．一只蚂蚁沿盒的外表面从点 A 处爬到点 B 处，那么它爬行的最短路程为_____．

13．如图，在长方形 $ABCD$ 中，$AB=4$，$BC=6$，分别以 B，C 两点为圆心，以 AB 的长为半径作弧，两弧在矩形 $ABCD$ 内部交于点 P，则点 P 到 AD 所在直线的距离为_____．

三、解答题（本大题共 5 个小题，共 48 分．解答过程写在答题卡上）

14．（本小题满分 12 分，每小题 6 分）

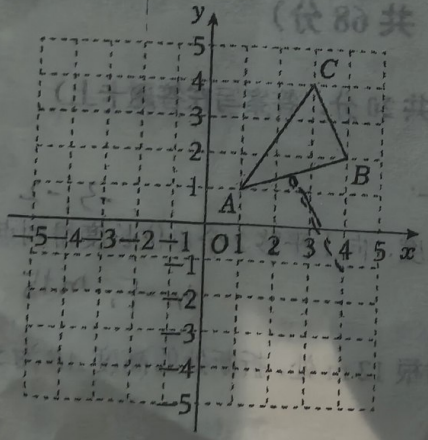
（1）$\dfrac{\sqrt{12}\times\sqrt{27}}{\sqrt{3}}-(\pi-3)^0-|-\sqrt{8}|+\left(\dfrac{1}{2}\right)^{-1}$　　（2）解方程组：$\begin{cases}2x+3y=17\\ \dfrac{x+y}{2}=y-2\end{cases}$

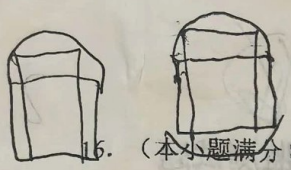
15．（本小题满分 8 分）如图，三个顶点的坐标分别为 A（1，1），B（4，2），C（3，4）．

（1）画出与△ABC 关于 x 轴对称的△$A_1B_1C_1$，并写出点 C_1 的坐标_____；

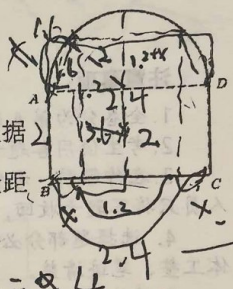
（2）在 x 轴上求一点 P，使得△APB 为等腰直角三角形，则点 P 的坐标_____；

（3）在 y 轴上存在点 Q，使得△ACQ 周长最小，画出点 Q 的位置，并直接写出△ACQ 周长的最小值（不写作法，保留作图痕迹）．





16．（本小题满分 8 分）如图，某单向单车道隧道的横截面由半圆和长方形构成，其中长方形的长等于半圆的直径．一辆宽 2.4 m、高 3.6 m 的卡车刚好能通过该隧道（车顶恰好接触隧道顶部），已知长方形的高为 2.0 m．



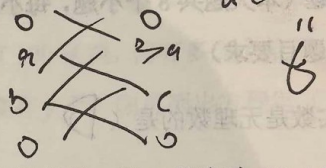
（1）求长方形的长；

（2）某物流公司有一辆宽 3.2 m、高 3.0 m 的货车需经过该隧道．根据交通运输安全规定，货车通过隧道时，车顶与该位置隧道顶部之间的安全距离不得小于 0.3 m．请问这辆货车能否安全通过该隧道？请通过计算说明．

17．（本小题满分 10 分）在平面直角坐标系中，O 为坐标原点，点 A 的坐标为（a，$3a$），点 B 的坐标为（b，c），

（1）已知 9 的平方根为 a 与 $2a-9$，b 是 $\sqrt{17}$ 的整数部分，$\sqrt{c-2}$ 与 $\sqrt{12}$ 是同类二次根式，求 A、B 点的坐标；

（2）若 a、b、c 满足 $\begin{cases}4a-b-c+2=0,\\5a-2b-c+4=0,\end{cases}$ 且△ABO 的面积为 6（O 为坐标原点），求 AB 的长和 a 的值．



18．（本小题满分 10 分）

在长方形 $ABCD$ 中，$AB=6$，$BC=3$．若将长方形折叠，使 D 落在边 AB（含端点）上，落点记为 E，这时折痕与边 BC 或者边 CD（含端点）交于 G，与边 AD 交于 F．

（1）如图 1，当 G 与 C 点重合时，求 AE 的长度；

（2）如图 2，当 $AE=2$ 时，求 FG 的长度；

（3）在研究上述问题的过程中，聪明的小张发现，无论如何折叠△DEG 都是一个等腰三角形．他进一步猜想：这个等腰△DEG 是否存在面积的最大值？若存在，请帮他求出此时 BE 的长度；若不存在，请说明理由．

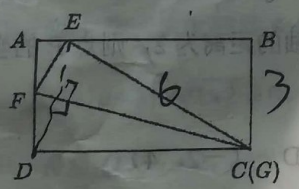


图 1

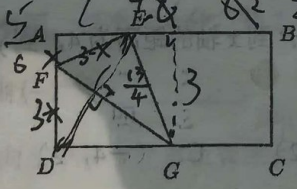
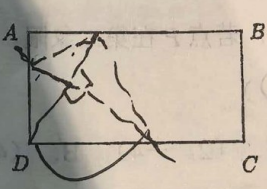


图 2



备用图

B卷（共50分）

一、填空题（本大题共5个小题，每小题4分，共20分．答案写在答题卡上）

19．比较大小：$\frac{1+\sqrt{5}}{2}$ ＿＿＿＿ 1.5（填“＞”、“＝”或“＜”）．

20．已知x，y是实数，且$y<\sqrt{x-1}+\sqrt{1-x}+1$，则$y+\sqrt{y^2-2y+1}=$＿＿＿＿＿＿．

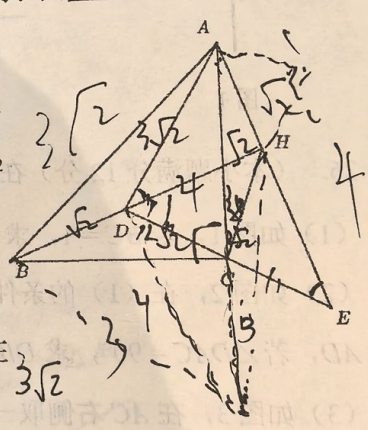
21．已知关于x，y的二元一次方程组$\begin{cases}x+y=2\\2x-y=m-1\end{cases}$（其中$-5\leqslant m\leqslant 5$）的解为整数，当点$P(x,y)$落在第二象限时，此时$x+2y$的值为＿＿＿＿＿＿．

22．定义：在$\triangle ABC$中，点P在BC边上，连接AP，若AP长为整数，且$\triangle ABP$的周长也为整数，则称点P为BC边上的“完美整点”．

（1）在平面直角坐标系中，已知点$A(0,3)$，点$B(-4,0)$，点$C(6,0)$，则线段BC上的“完美整点”的坐标为＿＿＿＿＿＿；

（2）在平面直角坐标系中，已知点$A(-7,24)$，点B、C在x轴上，且$\triangle ABC$中$AB=AC$，若底边BC上恰好有4个“完美整点”，则底边BC的长为＿＿＿＿＿＿．

23．如图，在$\triangle ABC$中，$\angle ACB=90°$，$AC=BC=3$，D为$\triangle ABC$内一点，连接BD，DC，延长DC到点E，使得$CE=DC$．连接AE，交BD的延长线于点H．当$BD=\sqrt{2}$，且$AB^2=AE^2+BD^2$时，则$AH=$＿＿＿＿＿＿，四边形$ADCH$的面积为＿＿＿＿＿＿．



二、解答题（本大题共3个小题，共30分．解答过程写在答题卡上）

24．（本小题满分8分）某校甲、乙两班共有100名学生，他们的体育达标率（达到标准的百分率）为81%．如果甲班学生的体育达标率为87.5%，乙班学生的体育达标率为75%

（1）．求甲、乙两班各有多少名学生？

（2）年级决定从甲、乙两班所有达标学生中组织一次体育展示活动，需要租用A型、B型两种客车．调研发现：A型客车每辆可坐9人，B型客车每辆可坐12人，要求恰好坐满，且两种客车都租用．A型客车每辆租金120元，B型客车每辆租金180元．问有几种租车方案？哪种方案最省钱？最少租金是多少？

25. （本小题满分 10 分）在平面直角坐标系 xOy 中，点 A 坐标为（$-\sqrt{3}$，1），点 C 在第一象限内.

（1）如图 1，连接 AC，交 y 轴于点 D，若点 D 坐标为（0，2），点 C 到 y 轴的距离为 $\sqrt{3}$，则 $AC=$_____；

（2）在（1）的条件下，在 y 轴上取一点 P，使得 $\triangle APC$ 为等腰三角形，请直接写出点 P 的坐标；

（3）如图 2，点 B 是 x 轴正半轴上的一动点，以 AB 为边作等边 $\triangle ABC$，连接 AO，当 $OB=OA$ 时，求 C 点的坐标.

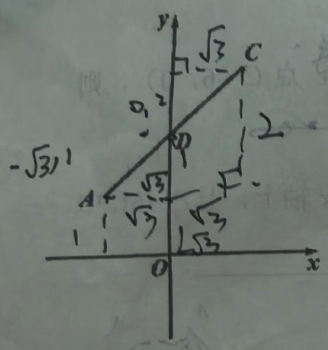


图 1

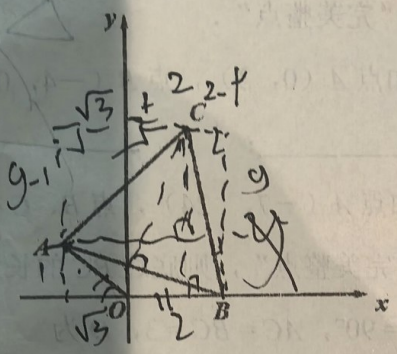


图 2

26. （本小题满分 12 分）在 $\triangle ABC$ 中，$AB=\sqrt{2}$，$\angle ABC=135°$.

（1）如图 1，若 $BC=1$，求 AC 的长；

（2）如图 2，在（1）的条件下，在线段 BC 右侧，作 Rt$\triangle BDC$，其中 $\angle DBC=90°$，连接 AD，若 $\angle DAC=90°$，求 DB 的长；

（3）如图 3，在 AC 右侧取一点 D，连接 BD、AD、CD，若 $\angle DBC=90°$，$\angle ADB+\angle ADC=180°$，试探究 $BD+DC-BC$ 值是否为定值，若为定值，请说明理由.

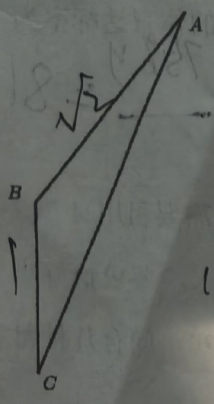


图 1

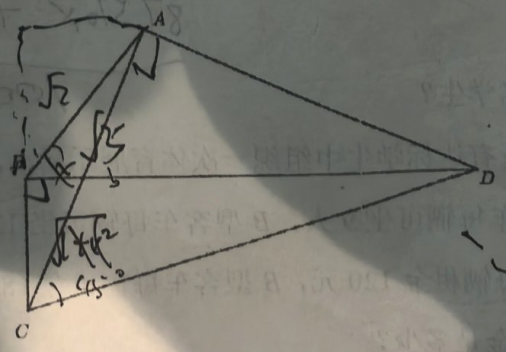


图 2

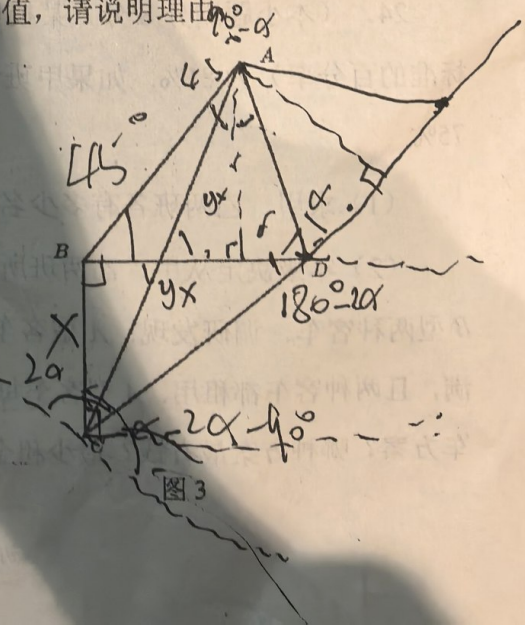


图 3